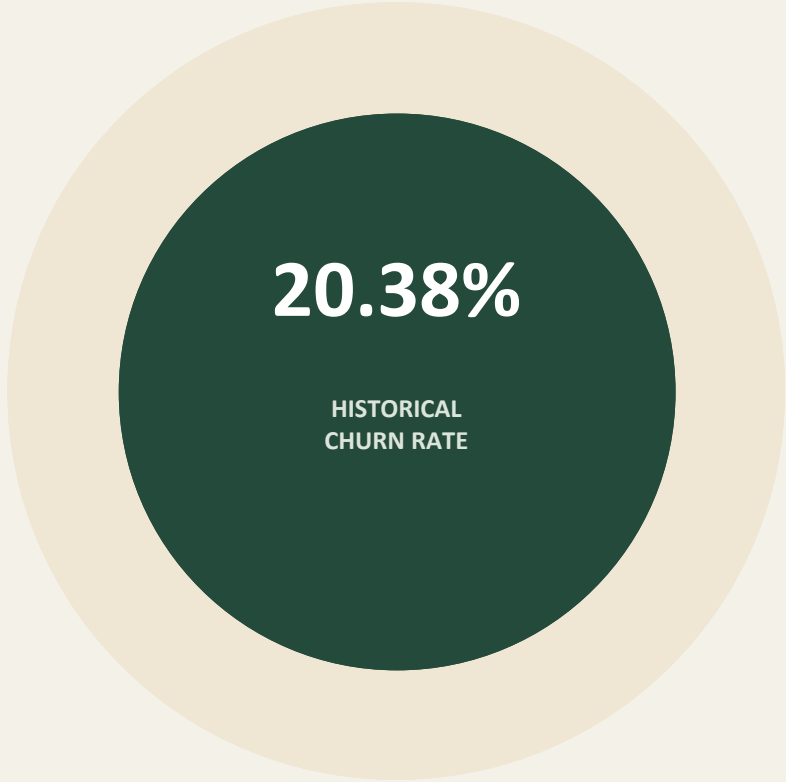
BANK CUSTOMER CHURN ANALYSIS

Where historical churn concentrates - and what retention teams should investigate first.

PORTFOLIO PROJECT

Data-led customer segmentation • Executive review



10,000
CUSTOMER RECORDS

2,038
CUSTOMERS EXITED

A 20.38% historical churn rate masks concentrated segment risk

The dataset identifies where a retention team should investigate first - not why customers left.

10,000

CUSTOMER RECORDS

18 source columns

2,038

CUSTOMERS EXITED

20.38% of file

20.38%

HISTORICAL CHURN

2,038 / 10,000

3

GEOGRAPHIES

France • Germany • Spain

GEOGRAPHY

Germany has 32.44% churn (814/2,509), about twice France and Spain.

AGE

The 50-59 band records 56.04% churn (487/869).

ENGAGEMENT

Inactive customers: 26.87%; active: 14.27%.

PRODUCT MIX

3-4 products show very high rates, but n=266 and n=60 respectively.

INTERPRETATION GUARDRAIL • Historical associations are not proof of causal drivers.

The business needs a sharper view of where to investigate churn

A focused segmentation view helps replace broad, untargeted retention activity with evidence-led diagnostics.

BUSINESS PROBLEM

The bank lacks a clear, evidence-led view of which customer profiles and engagement patterns are associated with account exit.

The analysis converts historical customer records into measurable retention segments and identifies where teams should investigate before designing interventions.

THE ANALYSIS SHOULD ANSWER

01 **Where should teams investigate first?**

Compare within-segment churn rates and population sizes.

02 **What customer patterns deserve review?**

Examine geography, age, product count, engagement and balance.

03 **What can be acted on responsibly?**

Separate observed association, hypothesis, cause and tested intervention.

The source file is complete for this cross-sectional EDA

10,000 records × 18 columns. The target is Exited (1 = exited, 0 = retained).

10,000	18	0	0	0
ROWS	COLUMNS	MISSING	DUPLICATE ROWS	DUPLICATE CUSTOMER IDs

ANALYTICAL WORKFLOW

01 / AUDIT

Schema, target distribution, nulls, duplicates and identifiers.

02 / EDA

Within-segment churn rates, counts and descriptive comparisons.

03 / TEST

Chi-square and Mann-Whitney tests; effect-size context.

04 / REVIEW

Static SQL logic review and Power BI screenshot audit.

SCOPE LIMIT • No date, event timestamp or defined lead time is supplied. Calendar trends, cohort churn curves and 30/60/90-day prediction cannot be supported.

Germany's churn rate is nearly double the other two markets

Germany has 814 exits on a base of 2,509; France has 811 exits on a base of 5,014.

Historical churn rate by geography



WHAT THE RATE REVEALS

814

German customers exited

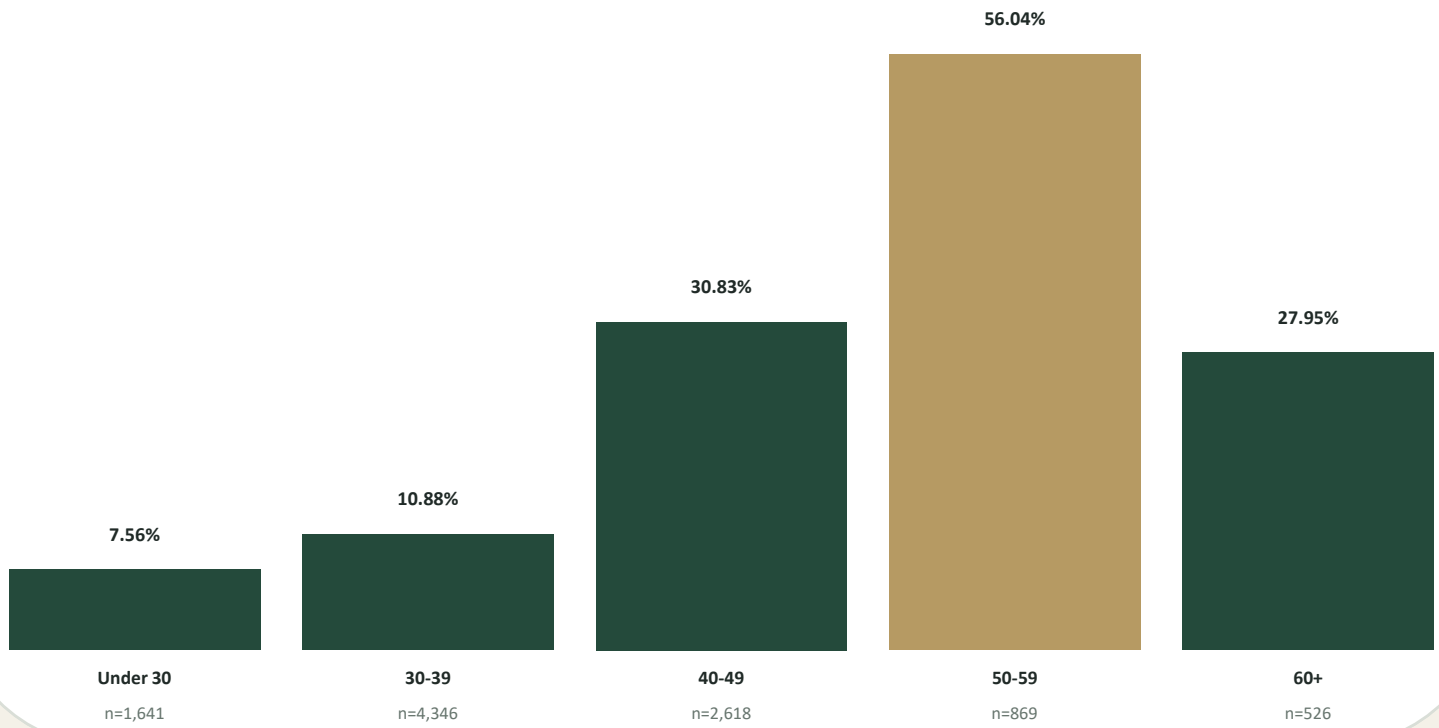
France records 811 exits despite a customer base roughly twice as large.

Next step: compare German customer mix by age, tenure, products, balance and activity before attributing the gap to a cause.

Churn peaks in the 50-59 age segment

The segment rate is 56.04% (487 churned out of 869); average age is also higher among exited customers.

Historical churn rate by age group



AVERAGE AGE

44.84

Years • churned customers

37.41

Years • retained customers

Interpretation: investigate service fit and product needs across older cohorts; age is an association, not a proven cause.

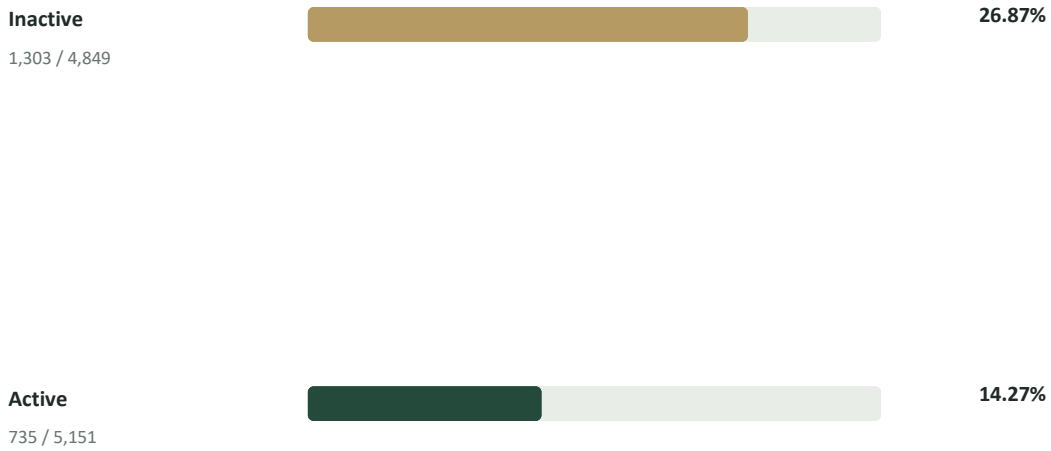
Product count is uneven; inactivity offers a testable signal

Always pair the extreme rates with their denominators before deciding where to act.

Churn rate by number of products



Churn rate by membership status



ACTION • Audit the 3+ product journey; pilot inactivity-based outreach against a comparable control group.

Balance and gender differences merit diagnosis; card type does not

The rates below are historical segment comparisons, not evidence of a causal mechanism.

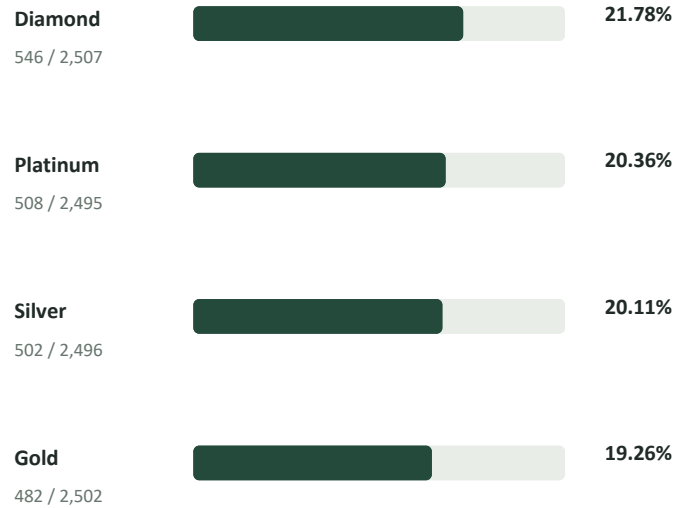
Balance group



Gender



Card type



STATISTICAL CHECK • Gender association: $p < 0.001$. Card type: $p = 0.168$ (not significant at 0.05). Avoid over-interpreting card-type differences.

Not every visible difference is a usable churn signal

Statistical significance, feature timing and sample design determine what can be interpreted responsibly.

NO SIGNIFICANT SIGNAL

Satisfaction score

Rates range from 19.64% to 21.80%; chi-square p = 0.433.

LEAKAGE WARNING

Complaint status ≈ outcome

2,034 of 2,044 complaint-flagged customers exited (99.51%); correlation with Exited = 0.9957.

MODEL / FORECASTING STATUS

Do not present the legacy model scores as validated results.

The attached EDA notebook explicitly excludes machine learning and has no saved outputs. Model scores in the separate brief are historical context only. No event timestamps or forecasting lead time exist in the CSV.

The current dashboard is close; tighten labels and encodings before submission

The screenshot is visually inspected; the underlying PBIX/DAX was not independently executed.



USER-SUPPLIED DASHBOARD SCREENSHOT

KEEP

- Geography slicer
- Gender slicer
- Card Type slicer
- Age Group slicer

FIX BEFORE SUBMISSION

- Rename the Diamond/Platinum/Silver/Gold panel to 'Churn Rate by Card Type'.
- Sort age bands chronologically.
- Replace rate donuts with bars or clarify their denominator.
- Show exact KPI values and add Reset Filters.

Move from segment discovery to controlled retention tests

Prioritize the actions by evidence strength, sample size and ability to measure incremental impact.

01	**Germany diagnostic**	Compare geography after adjusting for age, tenure, product count, balance and activity mix.	Retention + Regional Banking
02	**Multi-product journey review**	Audit onboarding, bundling and servicing for 3+ product customers; begin with a focused sample.	Product + CX
03	**Inactivity pilot**	Test re-engagement with a control group; separate customers who still hold a balance.	CRM / Retention
04	**Segment context**	Review balance and gender differences using product mix, service experience and customer feedback.	Analytics + Risk
05	**Data governance**	Establish event timing, leakage controls and cohort validation before predictive use.	Data + Model Risk

Measure incremental retention - not activity volume alone

The next decision should follow controlled evidence, not an untested segment label.

SUCCESS MEASURES

Retention impact	Churn change vs. a comparable control group
Targeting precision	Share of contacted customers who churn
Coverage	Churners reached at available outreach capacity
Efficiency	Churners identified per 100 outreach attempts
Commercial value	Retained value net of incentives and contact cost
Governance	Calibration, drift, fairness and segment performance

MANAGEMENT TAKEAWAY

Investigate.
Test.
Validate.

Investigate high-rate segments with customer-mix controls. Test interventions against a control group. Validate timing and data quality before any predictive deployment.

Evidence-led retention starts with the right questions - not the loudest chart.